

Cost

The main considerations in choosing a 3D printer are generally speed and cost. For the printed prototype, choice and cost of the materials, and colour capabilities are important.

Right now, the least expensive 3D printers are those using FDM technology. Entry-level printers cost ₹ 20,000 to ₹ 80,000. These use standard filament, can print one colour/material at a time, and have small build size. Most FDM printers use only PLA and/or ABS material.

Better printers cost ₹ 60,000 onwards. Some of these can print two colours/materials at a time; they usually print at finer resolution to get smoother surfaces. Moreover, the build size is larger, so you can print larger objects.

If you are good with your hands and tools you may buy a printer kit and assemble the printer by yourself. Prices of kits start at ₹ 6000 and go up to a few thousands. Bobs CNC RP9v2 Deluxe, Reprap Guru Prusa I3, HIC-TOP Auto Leveling Desktop, OneUp v2, HIC-TOP Desktop 3D Printer are all under ₹ 30,000 3D printer kits.

More expensive printers use SLA (stereolithography) or SLS (selective laser sintering) technologies. SLA printers use a laser or digital projector and photosensitive resins.

Prices, of course, depend on materials' cost, in-fill, size of the item to be printed and many other factors. Both proprietary and generic brand filaments are available in many different sizes and colours. If you wish to save money in the long run, a printer that accepts generic or

TABLE II
SOME IMPORTED PRINTERS AVAILABLE IN INDIA UNDER ₹ 100,000

Name of printer	Specifications
LulzBot Mini 3D Printer	• Manufacturer - LULZBOT • Maximum build area - 15.2x15.2x15.7 cm • Technology - FFF • Number of extruders - 1 • Number of colours - 1 • Highest resolution - 50 microns • Ports - USB 2.0 • Materials supported - ABS, PLA, Nylon, HIPS
MakerBot Replicator Mini Compact	• Manufacturer - MakerBot • Maximum build area - 9.9x9.9x15 cm • Technology - FFF • Number of extruders - 1 • Number of colours - 1 • Highest resolution - 200 microns • Ports - USB 2.0 • Materials supported - ABS, PLA
XYZprinting Nobel 1.0	• Manufacturer - XYZprinting • Maximum build area - 20x20x12.7 cm • Technology - FFF • Number of colours - 1 • Highest resolution - 200 microns • Ports - USB 2.0 • Materials supported - ABS, PLA
XYZprinting Da Vinci Jr. 1.0	• Manufacturer - XYZprinting • Maximum build area - 15x15x15 cm • Technology - FFF • Number of extruders - 1 • Number of colours - 1 • Highest resolution - 100 microns • Ports - USB 2.0 • Materials supported - PLA
XYZprinting Da Vinci 1.0 AIO All-in-One	• Manufacturer - XYZprinting • Maximum build area - 19.8x19.8x19.8 cm • Technology - FFF • Number of extruders - 1 • Highest resolution - 100 microns • Ports - USB 2.0 • Materials supported - ABS, PLA

cheap filament can help cut down long-term costs.

Frahan says, "3D printers today can be used to make everything from paper clips to jet engines. The same goes for market value because desktop FDM 3D printers are available under ₹ 100,000 while industrial ones can cost well over ₹ 100,000."

"The major advancement happening with respect to FDM is scaling down of the cost of 3D printers using this technology," informs Prudhvi Reddy, co-founder,

Think3D.

Nitin Gandhi, co-founder, LBD Makers, says, "For printers within this price range, recent notable additions have been Wi-Fi connectivity, auto bed levelling, Bluetooth connectivity and dual nozzles."

Sangani adds, "MaherSoft's core competency is to build high-quality FDM 3D printers at an affordable cost, making it more accessible to the masses. MaherSoft will be launching a crowd funding campaign on Indiegogo, where the Indie-Desktop 3D printer will be available for a pre-order price of approximately ₹ 30,000."

Will 3D printing live up to the hype

Despite its value in rapidly creating prototypes, 3D printing has very limited application as a production process. That is because 3D printing is intrinsically serial in nature; each unit of material has to be laid down sequentially. As a result, 3D printing is very slow relative to an intrinsically

parallel process like moulding, in which the material transformation happens simultaneously for the whole object.

For instance, an injection moulding machine in a factory can spit out 100 perfect plastic spoons every 15 seconds. The best 3D printers can produce one not-quite-as-good spoon in ten minutes, which is 4000 times less productive. So the plastic spoon you get at the ice cream shop will never be 3D printed.

There will, however, be two areas in which 3D printing will